Kaplan-Meier Estimation

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Introduction

The Kaplan-Meier (KM) estimator is the non-parametric maximum-likelihood estimator of the survival function from right-censored time-to-event data. Its signature step-function plot is the single most recognisable figure in clinical research. The estimator makes no distributional assumption about the time-to-event variable; it simply accumulates the conditional probability of surviving each observed event time.

Prerequisites

The reader should understand that a survival analysis studies time until an event, and that "right censoring" means some subjects are observed only up to a point before their event has occurred. Familiarity with R's `survival` package is not required but helps.

Theory

Let $t_1 < t_2 < \ldots < t_k$ denote the distinct event times observed in the data. At time t_j, let d_j be the number of events and n_j the number of subjects still at risk (not yet having had the event and not yet censored). The Kaplan-Meier estimator of the survival function is

$$\hat{S}(t) = \prod_{j:t_j \le t} \left(1 - \frac{d_j}{n_j}\right).$$

The estimator is a step function that drops at each event time by the fraction of the at-risk set experiencing the event. Censored subjects contribute to the risk set up to their censoring time but do not cause drops.

The standard-error estimator due to Greenwood is

$$\widehat{\mathrm{Var}}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{j:t_j \le t} \frac{d_j}{n_j(n_j - d_j)}.$$

Pointwise confidence intervals are typically constructed on the complementary log-log scale to ensure they lie in $[0, 1]$.

The **log-rank test** compares survival between two or more groups. It sums the observed-minus-expected events at each event time across groups, yielding a test statistic that follows a χ^2_{k-1} distribution under the null hypothesis of equal survival functions.

Assumptions

Kaplan-Meier estimation assumes:

1. Censoring is **non-informative** – subjects who are censored at time t have the same future risk as those who remain under observation. Violated when, for example, patients drop out because they feel worse.
2. Subjects are independent.
3. Risk of event does not depend on calendar time beyond what covariates capture (important when accrual spans a long period).

R Implementation

```r
library(survival)
library(survminer)

data(lung, package = "survival")

fit <- survfit(Surv(time, status) ~ sex, data = lung)

print(fit)

ggsurvplot(fit,
           data = lung,
           conf.int = TRUE,
           pval = TRUE,
           risk.table = TRUE,
           legend.labs = c("Male", "Female"),
           xlab = "Days from diagnosis",
           ylab = "Survival probability",
           palette = "Set2")
```

The `Surv()` call packages the time and event-indicator columns into a single survival object; `survfit()` with a right-hand side of `sex` produces separate KM curves for each sex. `ggsurvplot()` draws the curves with pointwise confidence bands, a log-rank p-value, and a risk table below.

Output & Results

The `lung` dataset has 138 male and 90 female patients with advanced lung cancer. The KM median survival is 270 days for males (95% CI 212 to 310) and 426 days for females (95% CI 348 to 550). The log-rank test gives $\chi^2_1 = 10.3$, $p = 0.0013$, indicating a significantly higher survival for females.

Interpretation

Reporting follows the CONSORT conventions: "Median overall survival was 270 days (95% CI 212-310) for males and 426 days (95% CI 348-550) for females; log-rank $p = 0.001$ (Figure X)." The figure should always be shown with a risk table beneath it – the curve is nearly meaningless once the number at risk drops below a handful.

Practical Tips

- Always include a risk table. Survival curves at late follow-up with only 2 or 3 subjects remaining can swing wildly and mislead readers.
- Do not rely on the median alone as a comparison statistic; if the curves cross, the median can flip direction as the study matures.
- The log-rank test is most powerful when the proportional-hazards assumption holds. For curves that cross or diverge late, consider the Renyi (weighted) log-rank or a restricted mean survival time comparison.
- Do not attempt a KM estimate without an event in the at-risk set; the curve will simply be flat at 1.
- When follow-up is short relative to the event rate, many subjects will be censored and the KM estimate at late times will have wide confidence bands. Report the cumulative event count at a clinically relevant landmark (1-year, 5-year) rather than the tail of the curve.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation